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Dear Prof. Akira S Mori,

We would like to begin by thanking both you and Reviewer 2 for additional creative and valuable thoughts. On behalf of myself and my coauthors, we appreciate your time and contributions.

All reviewers provided carefully considered comments and we have done our best to accommodate those suggestions here. The most significant changes include:

1. a new discrete trait analysis of substrate usage that improves upon previous approaches.
2. a more careful GLM modelling approach that provides greater insight into our claims of morphological intermediacy being linked to niche breadth.

We believe these additions substantially improve our argument and the quality of the manuscript.

Below please find **reviewer comments in green**, and our responses in black. New text added to the manuscript is **highlighted**. We have formatted a page break between editor and reviewer comments. Where necessary, we have interjected our responses among reviewer comments.

On behalf of myself and coauthors, we thank you for your time and consideration,

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Editor:

As you can see, reviewer 2 is happy with the new models that you and they have co-created to test some of the major predictions in this paper, particularly linking generalism and a species' position in morphospace. And, as you can see, reviewer 2 found some mis-characterizations of the support for other predictions that need to be addressed.

The study remains very rich and I expect will be of interest by ELE readers, even if many of the specific tests are not supported statistically. Highlighting the few that are will be important, of course.

We appreciate the energy and attention that everyone has invested in our manuscript. This has been a truly productive review process and we feel like the contributions of you and the reviewers have introduced new methods and ideas that strengthen our arguments. Thank you for your support of our manuscript throughout this process. We take this all as an encouraging sign as we address all outstanding comments below.

Here are my (independent of R2's) comments on this version:

new line 83: not a sentence.

We have amended this phrase to form a complete thought/sentence.

line 88: the landscape is measured using the ecologies and morphologies (measured or reconstructed), so this sentence seems circular.

Yes, this sentence was conceptually redundant. We have adjusted it to better convey our interest in how the landscape has changed through time and how that has been reflected in the ecology of the organisms it represents.

Importantly, we can ask which axes have shaped the landscape at different points in time and how these changes are linked with shifts in ecology, habitat, and/or geographic range.

104; "unique" or "varied"?

Definitely "varied". We have changed this.

107: this seems like it is referring to line 81, but it is not clear. In other words, this "idea" comes a bit from nowhere.

We agree that the reference to the hypothesis previously presented (line 81) is too disconnected. We have updated the text to instead reintroduce the idea.

Here we present and test the idea that specialist peaks in the adaptive phenotypic landscape are linked through ancestors that were morphologically intermediate habitat generalists.

107++: I think you need to say you also score the species "generalism" score based on habitat use here, since that is key to the paper.

We agree that our idea of "generalism" needs to be linked to habitat usage here, but think that the introduction is not the appropriate place to specify exactly what the data are. We have attempted to find a balance by mentioning that generalism is related to habitat. This is presented in the comment above.

114-115: large trait changes will perforce lead to new morphologies, so this sentence seems a bit circular. Rework

The idea we are trying to present is that the heterogeneous evolutionary process of some traits means that instead of gradual changes we can rapidly accumulate big trait differences. We have adjusted this sentence to capture that.

117: given the strength of evidence, perhaps "we reconstruct a scenario where..." rather than "we identify"?

I think in all science we are "reconstructing scenarios" given the collected data, and I do not think that changing the wording from "we identify" provides a more nuanced reflection of the strength of the evidence.

239: "To test the idea that..." this is vague. Can a better prediction (either from first principles or from past work) be substituted here? (This concern is also raised by R2.)

We now provide context for investigating the correlation between ecology and morphology by referring to important review works on the field of ecomorphology.

Ecomorphology predicts a correlation between the ecology and morphology of organisms (Bock 1994; Ricklefs & Miles 1994) and underpins the idea of adaptation. To test the idea that habitat usage in dragons is dictated by morphological position ...

285, 288: AICw of 0.50 is non-existent statistical support for these claims. Can you do a likelihood ratio test here?

We have carried out some changes to our discrete model fitting exercise that make our approach more rigorous and present uncertainty more transparently. I will go into detail here to explain exactly what we did.

The first change was to correct the scoring of a single species *Ctenophorus yinnietharra*. This correction was brought to our attention by an eagle-eyed reader of our preprint. They pointed out that this species is a rock-specialist of decomposing granite outcrops in Western Australia, and should be coded to reflect that (changed from 'generalist' and '3 substrates' to 'specialist 3' and '1 substrate'). This is worth explaining, because this is a species embedded within a clade of narrow-breadth niche specialists. Its prior coding forced a transition from an embedded specialist state back to a generalist state, resulting in model fitting and ancestral state estimation artefacts. Recoding this taxon made it apparent that transitions from terrestrial

taxa to rock living taxa have occurred more than once, and should be factored into the model fitting exercise.

Perhaps more importantly, this made us reconsider how we have been modelling the discrete substrate traits. In both current cases (4-State **Generalist-Specialist**; 5-State **Niche-Breadth**) we attempted to identify the paths through which species change ecologies by looking at how observed usage can be categorized (Generalist vs. Specialist) or by looking at the number of discrete categories used (Breadth). Both exercises required aggregating and recoding the data, moving it one step further away from the original state scores. This practice may inadvertently introduce imbalance or bias. To address this shortcoming and tackle both evolutionary questions simultaneously (evolution of niche breadth, paths between generalist/specialist) we reassessed our modelling approach.

We think we can capture both questions together by modelling the acquisition and loss of individual trait states using a polymorphic state approach. This should better reflect the observed data because it directly models species' traits instead of further reclassifying them. For example, *Intellagama lesueurii* is scored for 'aquatic', 'terrestrial', 'low structure', 'high structure', and 'arboreal' substrate states. In the **Niche-Breadth** exercise this species was recoded as breadth = 5', and in the **Specialist-Generalist** and as 'generalist' in **Specialist-Generalist**. In our updated approach, *I. lesueurii* is coded as G+L+H+T (we have dropped the 'aquatic' state because it is rare and to limit the model parameter space). Modelling the actual character states and not summaries of those states provides nuance about where and when individual substrate states were potentially lost or gained. Incorporating polymorphism also allows us additional flexibility in modelling transitions. We can test more informative pathways such as ordered transitions between adjacent or semi-adjacent substrate states. We fit models using the *fitpolyMk* function of *phytools*. As a result, we find support for two rate, ordered models (transitions between adjacent states occur through the acquisition or loss of polymorphisms), in which specialisation (loss of a substrate state) occur at nearly nine times the speed of increasing generalisation (acquisition of a substrate state). While we view our new modelling approach as an improvement over our previous approaches, we have retained the **Niche-Breadth** and **Specialist-Generalist** modelling in the supplement for transparency.

In addition to this new modelling approach, we have also addressed how we compare models and use them to estimate ancestral states. Because not all models designed are nested, we cannot be directly compared via likelihood ratio tests. We address this challenge in two ways. (1) We provide delta AIC values as part of Figure S13 which shows the model structures, AIC values, weights, and (now) relative fits. (2) To account for perceived lack of confidence in any one model, we employ a model-averaging approach to reconstructing ancestral states. After fitting our models and determining AIC weights, we integrate all fitted models into the ancestral state estimation by accounting for them in proportion to their AIC weights. We present ancestral state estimations under all competing models with $\Delta AIC \leq 2$.

All changes to our discrete trait modelling, including methods, results, and discussion are integrated directly into the manuscript.

317: likewise, I think it would be better to say that there is no support for any particular ancestral state using these particular methods. This is not surprising for something that evolves at a high rate (many transitions) with many states.

We agree that even in the best of circumstances, it can be hard to confidently estimate ancestral states. To more transparently report ancestral estimates across competing models, we have added a step that uses model averaging to reflect uncertainty. We describe this more in our response above and in the text.

327: You do need to report the pattern from PC2 here.

To expand our GLMM model fitting we included additional predictors (previously PC 1-3 ~82% of variance, now PC 1-6 ~92% of variance). Inclusion of additional predictors (and a change to a more appropriate zero-truncated Poisson model class) provide new results, including the significant influence of a linear relationship with PC 4. To correct our GLMM model reporting, we have expanded our results to include effect estimates (β), standard error (SE), Z scores (Z), and p-values estimated by the Wald Z test. To limit clutter in this section, we direct readers to the supplement to inspect parameter estimates for non-significant predictors.

PC 1 ($\beta=-1.09$, SE=0.42, Z=-2.55, P=0.010) shows a negative quadratic function where substrate breadth is highest at intermediate PC scores. PC 4 ($\beta=2.17$, SE=0.86, Z=2.53 P=0.11) shows a positive linear trend with substrate breadth. PCs 2, 3, 5, 6 do not show significant effects (see Supplement for statistics).

344: This statement (also in the Abstract) confuses me. Given this is a radiation and there is a lot of morphological difference among species, this is tautological: you are studying this radiation because it is diverse, and a diverse radiation will show a "marked expansion of morphological space." Because you are not comparing this to its sister clades, you cannot say that is particularly "marked." At 473, you cite papers that have already documented that the variation is larger than in other Iguanid clades.

While we appreciate your thoughts here, we disagree with the idea that the observed patterns are not comparative. The "marked expansion of the morphological landscape" is a comment on the diversification of Australian dragons relative to their relatives in Southeast Asia (*Physignathus*) and New Guinea (*Hypsilurus*, *Lophognathus*). Those other groups have done comparatively little to stretch the morphospace and generated only modest species richness (*Hypsilurus*—18 spp.; *Lophognathus*—3; *Physignathus*—1) despite their age. To support our claims that the Australian amphibolurinae are noteworthy, we reference two great papers by Gray et al. that look across Iguanians. We do not feel that results presented by Gray diminish our findings, because her work focused exclusively on skulls—just a single major axis of differentiation among these species.

357-374: this information can go in a more Herp-oriented biogeography paper to save space.

We appreciate your suggestions here. Through previous rounds of review we have already reduced this angle of our discussion from 3 paragraphs to 1.5. We have edited it further to fit within the word limits of Ecology Letters, but believe it is important that the temporal and phylogenetic context of the group is presented in the main text. For that reason, we have retained this section, but reduced it to a single more concise paragraph (reproduced below).

Phylogenomics of the Amphibolurinae

Dragon lizards (Agamidae) are a charismatic and diverse group of ~600 species found across Africa, Asia, and Oceania. Our new sequence-capture dataset extends our phylogenetic understanding of the primarily Australo-Papuan subfamily Amphibolurinae (Macey et al. 2000; Melville et al. 2001, Hugall et al. 2008; Shoo et al. 2008; Melville et al. 2018; Chaplin et al. 2020; Fenker et al. 2024). Amphibolurines likely originated in Late Cretaceous Asia before dispersing into Australo-Papua (Scarpetta et al. 2026). This biogeographic pattern is consistent with other Australian squamate groups (Keogh 1998; Esquerre et al. 2020; Brennan et al. 2021), though the long amphibolurine stem lineage is unique. The subsequent rapid Miocene diversification of the group, followed by a later slowdown is consistent with theory around adaptive radiations and dispersal into new regions (Harmon et al. 2010; Schluter 2000; Simpson 1944; Yoder et al. 2010).

377: I think it is safer to say that the "observed morphospace" is patchy, etc., and then say that we can think of that space as an adaptive landscape.

We have changed this in the main text.

382: Best to reiterate the statistical evidence for these intermediate morphologies facilitating transitions among peaks here, rather than just stating it. At 387, the link is called "theoretical."

We have expanded this paragraph to provide context to where these claims come from, so readers can more easily assess our claims.

The morphological landscape of amphibolurine dragons is dispersed and patchy, with peaks that largely correspond to recognized genera. Obvious phenotypic peaks relate to strongly arboreal (*Hypsilurus*, *Lophosaurus*) and terrestrial species (*Tympanocryptis*, some *Ctenophorus*) (Fig.1B). However, intermediate forms that use a variety of substrate types (*Amphibolurus*, *Gowidon*, *Lophognathus*, *Tropicagama*, *Pogona*) fill the morphological space. Evidence from GLMMs link these morphologically intermediate taxa to wider habitat usage and broader niches than more morphologically extreme relatives. This pattern, paired with evidence from discrete evolutionary models that indicate specialist niches evolve from generalist ancestors. This provides a pathway to unite disparate peaks of the landscape and facilitate transitions to new ecologies.

And just to follow-up on a confusion re. Fig.4: a "paradigm" is a set of beliefs or an archetypal model, neither which a clade can have, so clades cannot have "paradigm shifts."

We have amended the legend to Figure 4.

Referee 2:

The authors conducted the final analysis to lend numerical support to their verbal arguments. They have also partially added evidence of statistical support for some of these analyses, though they need to do so more clearly for a reader to actually be able to evaluate what they did. Now that the authors have more clearly presented these statistics it's clear where some mistakes in model analysis and interpretation have cropped up. These errors will need to be fixed, with resulting ramifications for some of the interpretations in the results and especially the discussion. Particularly, what is the biological meaning that only PC1 (when controlling for phylogeny) has a quadratic effect. Does this simply mean that intermediate body sizes are most likely to be generalists, and other traits matter less for generalization? Or are other traits at play in PC1?

We appreciate your considered thoughts. The topic of the biological meaning of the GLMM results is an interesting one. In our previous submission we struggled to incorporate these results without exceeding the journal word limit considerably. Figure S15 provides some context for how this result may be interpreted. We presented the PC loadings directly above the GLMM results in item B so that readers might be able to consider our interpretations. We agree that PC1 is largely driven by organism size. This is reflected coarsely by the largest dragons being arboreal and the smallest being terrestrial, with a variety of intermediate forms using multiple substrates. However, other traits do load heavily on PC1, including tail length, pelvic width, and body diameter. Again these are traits that are typically correlated with ecology: specialised arboreal dragons have laterally compressed bodies (low pelvic width and body diameter, long tails) and obligately terrestrial taxa have wider bodies (high pelvic width and body diameter) which they flatten against the substrate.

We have included these observations into the discussion:

Linking morphology to ecology however, is not always so clear. GLMMs provide some evidence that morphology is predictive of substrate usage. Along PC1 (47.9% of variance) intermediate morphologies show greatest niche breadth (Fig.5). This PC displays strong loadings for size, as well as tail length, and body width. These traits seem to partition across the arboreal and terrestrial axes. However, the linear relationship of PC4 (4.4%) with breadth highlights an interesting interaction with PC1. Species with longer limbs and heads (PC4) show a greater propensity for niche breadth irrespective of their size (Fig.5).

In summary:

Report evolutionary rates model comparison in a way that is transparent regarding the uncertainty and relative effectiveness between models. AIC weight not a good option. Delta AIC or null hypothesis testing (both of which indicate the equal rates model can't be rejected) is more transparent.

While we disagree that AIC weight is not a worthwhile measure of model comparison, we have expanded our approach to better reflect uncertainty in model choice (inclusion of delta AIC) and particularly ancestral state estimation (through model averaging). We discuss these at greater length above in response to the Editor.

Be clear regarding which GLMM models are used in main results, and report significant quadratic effects clearly and consistently.

Ensure that GLMM models are consistently run—do not switch between poisson and gaussian glms in the same analysis without confirming that one is better. That the gaussian actually has a lower AIC suggests that using the poisson is not in fact a very good model. On reflection, this is likely because there are no zero habitat breadths in the data (zeros would be a common result in a poisson distribution). Should try rerunning with a zero truncated poisson family, and otherwise ensure that model assumptions are being met through standard model validation practices.

Thank you for the suggestion of fitting using a zero truncated Poisson family. Conceptually it makes far more sense than the traditional Poisson. Following your advice, we tested several model families (Gaussian, Poisson, zero-truncated Poisson, truncated Conway-Maxwell Poisson) and assessed family/model fit and adequacy via three common tests (K-S, Dispersion, Outlier) in *DHARMa*. Our data best fit the assumptions of the zero-truncated Poisson family model and so we applied this to both exercises.

Adjust discussion in light of non-significant results.

First, I thank the authors for completing the analysis to show clear evidence that the center of morphospace is really the section of morphospace that leads to greatest habitat generalization (with the possibly strange exception of PC2, though non-significantly when phylogeny is considered...). This makes the argument hold together much better. Some additional technical information should be provided in the main text regarding the statistical outcomes of this model, to allow the reader to confirm that there is indeed a signal in support of this core argument without having to flip to the supplement (parameter estimates, Ses, test statistics, p-values) for each of the morphological axes tested. For example, it's not clear what model you're basing your statements on in the maintext. Given that you don't mention that PC2 has a significant effect I assumed you were using the phylogenetic model. But then PC3 isn't significant either, so why did you mention it? If you are using the non-phylogenetic model the quadratic for PC2 is also significant and should be mentioned and explained, since it has curvature that goes against your hypothesis—falsifying your hypothesis for at least one core morphological axis and suggesting multiple ways (in this morphological dimension) for becoming a generalist. If you don't discuss this point it has the appearance of sweeping counter evidence under the rug. However in the supplement figure S15 legend it says the plots are for the phylogenetic model (even though that one has a higher AIC than the non-phylogenetic model). But if that's what you're using to present data then neither PC2 nor PC3 has significant quadratic effects, and only PC1 has evidence of an effect. Statistical support for and against biological claims needs to be clearly presented. Here it is not.

We had no intention to sweep counter evidence under the rug, and apologise for any lack of clarity. The addition of multiple new analyses and the requests that their methods be included in the main text have pushed us above the word limit. We have done our best to address these critiques. Apologies for the inconsistent reporting on GLMM model fits. Given the phylogenetically structured shape of our data, we believe it is important to include the phylogenetically corrected models and prefer these. Inclusion of the non-phylogenetic

GLMMs was exclusively to satisfy requests in the previous review. We have corrected the text to reflect results of only the phylogenetically corrected models. As mentioned in response to the Editor, we expanded the number of predictors to include PCs 4-6 and applied the zero-truncated Poisson distribution. This is relevant because we find evidence for a significant effect of the quadratic effect of PC1 and a linear effect of PC4. We now report these results more consistently in the results, and address their importance (including their interaction) in the discussion.

Line comments:

L253: If space is needed (which the authors invoked to justify not make clarifying changes in the methods/results/discussion) this paragraph can largely or entirely be moved to the supplement. This information is insider baseball appropriate for a herp or taxonomy journal, but doesn't make sense to present front and center for the audience of ecology letters.

We understand the request, but would prefer to keep this information. Ecology Letters is an exceptional journal because it includes both theoretical and empirical work. It's the marriage of the two that first motivated us to submit to this journal, and so it would be a shame to lose elements of the empirical group that initiated this project.

L285: AICweights don't provide any meaning unless the full set of models that the weights are up against are shown. It would be more consistent to show either deltaAIC values between the best and next best model, or likely more relevant to show the p-value based on a likelihood ratio test between the model with the core hypothesized pattern (a model with one set of rates for transitions to/from generalists and another set of rates for transitions between specialists). Reading that only 50% of the AIC weight is in favor of the hypothesized model looks really bad, since it means that around 50% of the weight is in favor of some other sets of models! Looking at the supplement the generalist-specialist model (AIC 139.2) has support that's basically the same for an equal rates model (AIC 140.9). Even the most lax AIC considerations say that a model within 2 AIC points are equally good at explaining the data. There's really no indication here that there's good support to believe that specialist-generalist evolution is any more likely than specialist-specialist evolution. The results and discussion should be modified with this in mind, to acknowledge that there's maybe a trend in this direction, but there is certainly no firm evidence. Some of this may simply be due to lack of power. The MLEs for rates seem to suggest that the rates from generalist-specialists are 3X faster than between specialist-specialist, but even with this fairly dramatic difference in rates there still aren't enough transitions (apparently) to detect a difference in model fit. Using the AIC values above I back calculated the p-value based on likelihood differences between models and got $p = 0.17$. This p-value underscored to me that the data are not providing enough evidence to reject the equal rates model. I never the less think it's somewhat suspicious that the only evident specialist-specialist transition that happens in the phylogeny is the ground->rock transition which species are apparently able to do without passing through a generalist phase (based on figure 2).

There are a number of really valuable points included in this comment. We have addressed some above (in response to both you and the Editor). First, while we do think AIC weights provide sufficient context to compare models, we have also now included delta AIC values that will allow readers to compare models more quickly. More importantly perhaps, to reconstruct ancestral state estimates, we have changed to a model-averaging approach. This

method accounts for model weights in the marginal reconstruction of ancestral states, allowing us to reflect the uncertainty of competing models in our estimates. Further, we have extended our model fitting approach to include (what we believe) is a more appropriate discrete trait evolution model that incorporates state polymorphism to better track state transitions. Despite these additions, there are limited changes to the greater inferences we have made.

L330: Need to provide information on what the numbers are in the parentheses. I assume these are slope terms, but it's unclear. Not clear how p-values were calculated: provide test statistic.

We have expanded how we report the results of the GLMM exercise in the main text.

PC1 ($\beta=-1.09$, $SE=0.42$, $Z=-2.55$, $P=0.010$) shows a negative quadratic function where substrate breadth is highest at intermediate PC scores. PC4 ($\beta=2.17$, $SE=0.86$, $Z=2.53$, $P=0.11$) shows a positive linear trend with substrate breadth. PCs 2, 3, 5, 6 do not show significant effects (see Supplement for statistics).

L331: Need to show evidence that a quadratic model fits the data better than a linear model (i.e. parameter estimates, SEs, and P-vals), to demonstrate to readers that there is actually evidence here.

Please see our response above. We also provide the output of model fits in the Supplementary Material for readers to investigate, and the actual model fitting objects in the Supplementary Data.

L388: "Niche breadth modelling supports this by preferring pathways of narrowing substrate usage and favoring transitions between specialist and generalist states over transitions from one specialist to another (Fig.2)." Now that AIC weights have been provided this statement does not seem to be supported by the data (see comment above). There seems to be no ability to say that the generalist-specialist model is supported over the equal rates model ($p = 0.17$ based on my calculation above). This phrasing, and the conclusion throughout the discussion should be toned down accordingly.

We agree with this sentiment. We believe the new model fitting analyses we include better address both questions presented in the previous modelling exercises and better reflect the evolutionary gain and loss of habitat usage.

L393: "Specialist forms often descend from morphologically intermediate ancestors, and in some cases are nested deeply within generalist clades." The word 'often' is carrying a lot of weight here, given that the notable exception is that rock specialists repeatedly evolved from ground specialists without passing through a generalist intermediary.

Supplement:

GLM model fitting. Define variables. Unclear what 'mah' is based on reading the supplement.

We have expanded the materials in the supplement to better describe these details.

I am confused why there is a random effect of species, given that each species I believe only has 1 data point associated with it—explain the logic here. Perhaps doing so is to account for overdispersion without running a negbinomial, but this analysis decision would need to be justified. What is variable 'g', and what is 'agam.mat'—define all variables so models are interpretable. Is the variance structure equivalent to brownian motion or pagel's lambda? Or a different phylogenetic correlation structure? Why does the error structure switch from poisson for a count variable (which my understanding was that breadth was a count variable), to gaussian for the phylogenetic model? Do the residuals from the gaussian model run look alright? They presumably wouldn't if we're dealing with a count. Though the poisson itself may not be a very good fit because you're dealing with zero truncated data—looking at the plots it's clear that there are no species with habitat breadths of zero. Some serious kicking of the analytical tyres is needed here.

Thank you for your close assessment of the model properties. The random species effect should not have been included, as you correctly point out, we have included just a single representative per species. We have removed this random effect in our updated analyses. We provide a definition of all model variables in the supplement now. The variance-covariance matrix was generated under the assumption of Brownian Motion.

In the second set of models all are poisson. Though the fact that the gaussian model above has an absolute lower AIC than all other models makes me wonder if a gaussian is a better way to model these data than the poisson. Residuals need to be checked and model validation conducted. Note, given the data consider using "truncated_poisson()" as the family instead of either poisson or gaussian.

The inclusion of the model under a gaussian assumption was inappropriate and in error. Previously we fit all models under a Poisson distribution, but have since found that the zero-truncated Poisson is a much more appropriate fit. As evaluated by visual investigation of residuals and common tests of adequacy (K-S, Dispersion, Outlier) in *DHARMa*.

Unclear from figure s13 what "S1" "S2" and "S3" are. Supply this information in figure legend.

Though we have removed this model set from the main text we have retained it in the supplement and made sure to include an appropriate legend in Figure S13.

I like the analysis with the angles, though I think the color scheme is reversed when comparing S15C and 15D. The conclusion that early in the tree branches tend to move more towards the center than later in the tree is not obvious by looking at figure D (the phylogeny)—so finding a way to summarize that trend that's more visually obvious would be helpful (e.g. plot average branch angle as the y and node height on the x).

You are absolutely right, we accidentally flipped the color scheme in the legend, apologies. We have now fixed this. We also like the idea of plotting branch angle as a function of node height. We have carried this out and fitted a simple linear model to the data to show that there is a significant effect of node height on direction of change, indicating that older nodes are evolving towards the morphological center.